

Ensemble Streamflow Forecasting Using Deep Learning: A Case Study of Tropical Cyclone Helene

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Background

- Tropical Cyclone Helene caused severe flooding in North Carolina in September 2024, with 20–30 inches of rainfall over three days. The French Broad River at Asheville and the Swannanoa River at Biltmore reached peak stages of 23.1 ft and 27.33 ft, respectively
- At least 63 river gauges exceeded their historical peak flood levels during the event (National Hurricane Center, 2024).
- Nearly 30% of these gauges had missing observations during the peak of the event.

Approach

- Use streamflow observations from gauges across the southeastern U.S. and apply kriging to produce spatially and temporally continuous streamflow estimates during the storm.

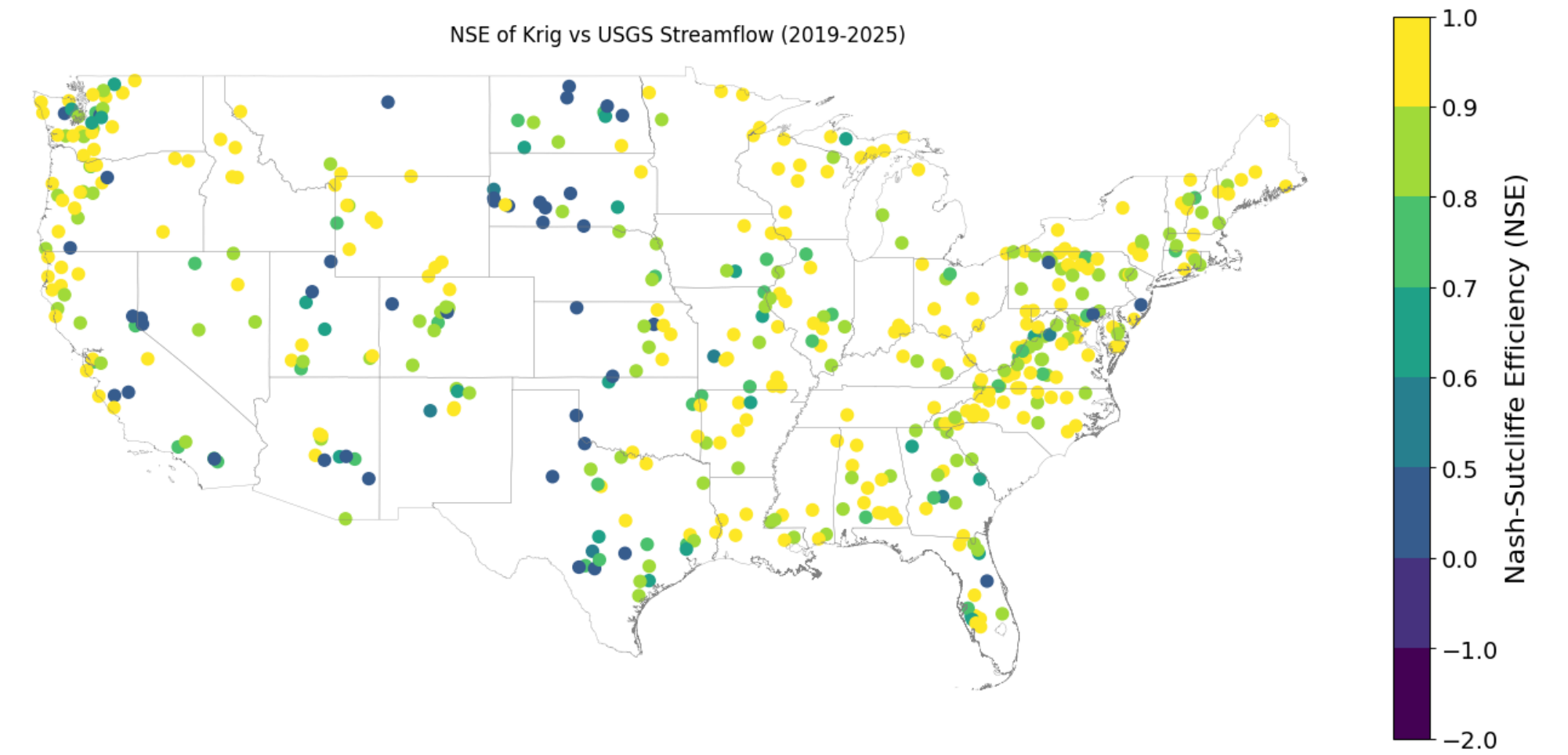


Figure 1 Nash Sutcliffe Efficiency (NSE) of daily kriged streamflow from k-fold cross validation at CAMELS sites

- Median NSE of 0.90 demonstrates the feasibility of spatially complete, continental-scale streamflow estimation and forecasting using kriging

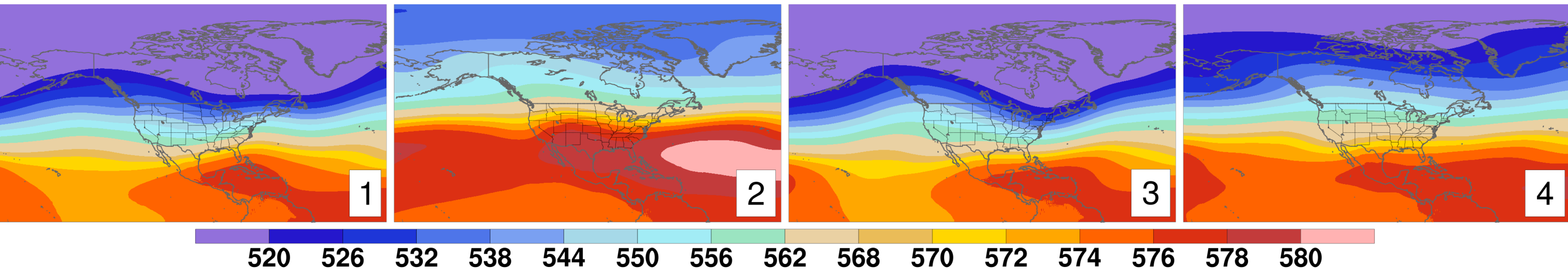


Figure 2 ERA5 geopotential height in dam across four CONUS meteorological regimes

- Kriged estimates maintain stable high performance across meteorological regimes. Median NSE Overall: 0.9, Regime 1: 0.88 (cool season), Regime 2: 0.85 (warm season), Regime 3: 0.89 (cool season), Regime 4: 0.90 (transition season)

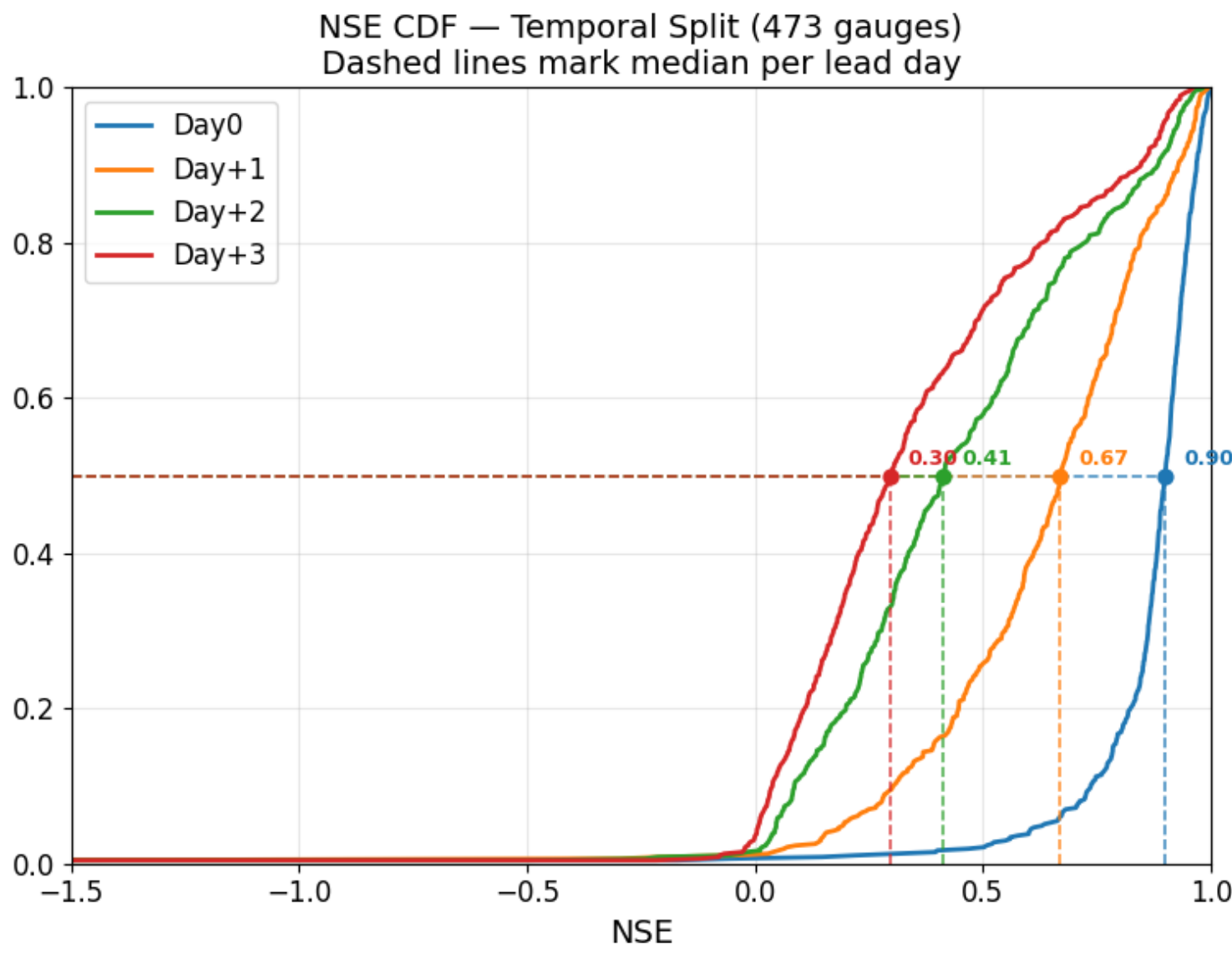


Figure 3 Distribution of NSE with lead time for daily streamflow forecasted using spatial temporal autoregression

- A convolutional neural network model was trained using daily kriged streamflow fields
- When initialized with six day ahead kriged streamflow, the CNN skillfully forecasted streamflow at gauged and ungauged locations with a nowcasting median NSE ~ 0.9

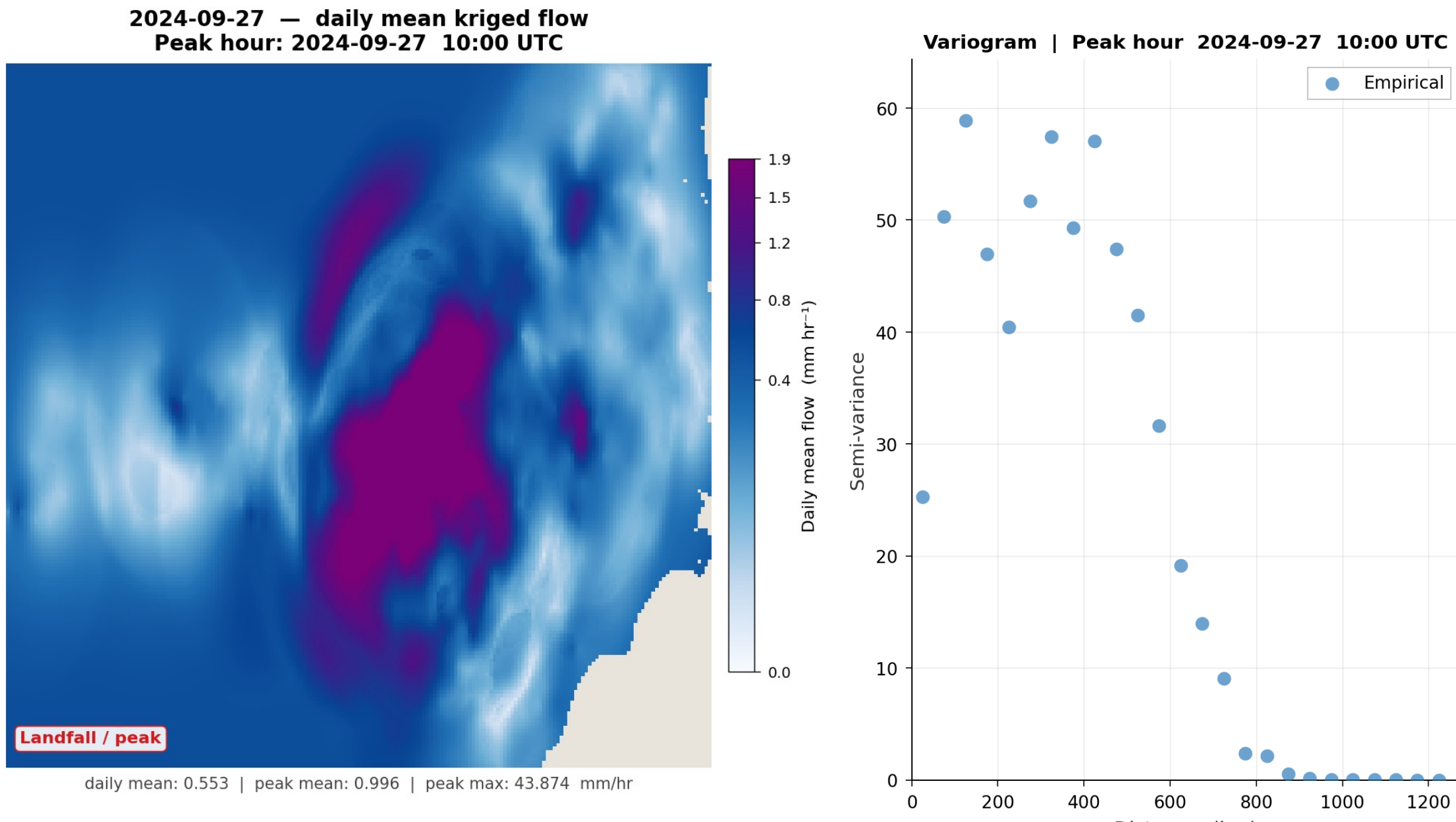


Figure 4 Kriged streamflow and empirical variogram at the peak hour during Helene

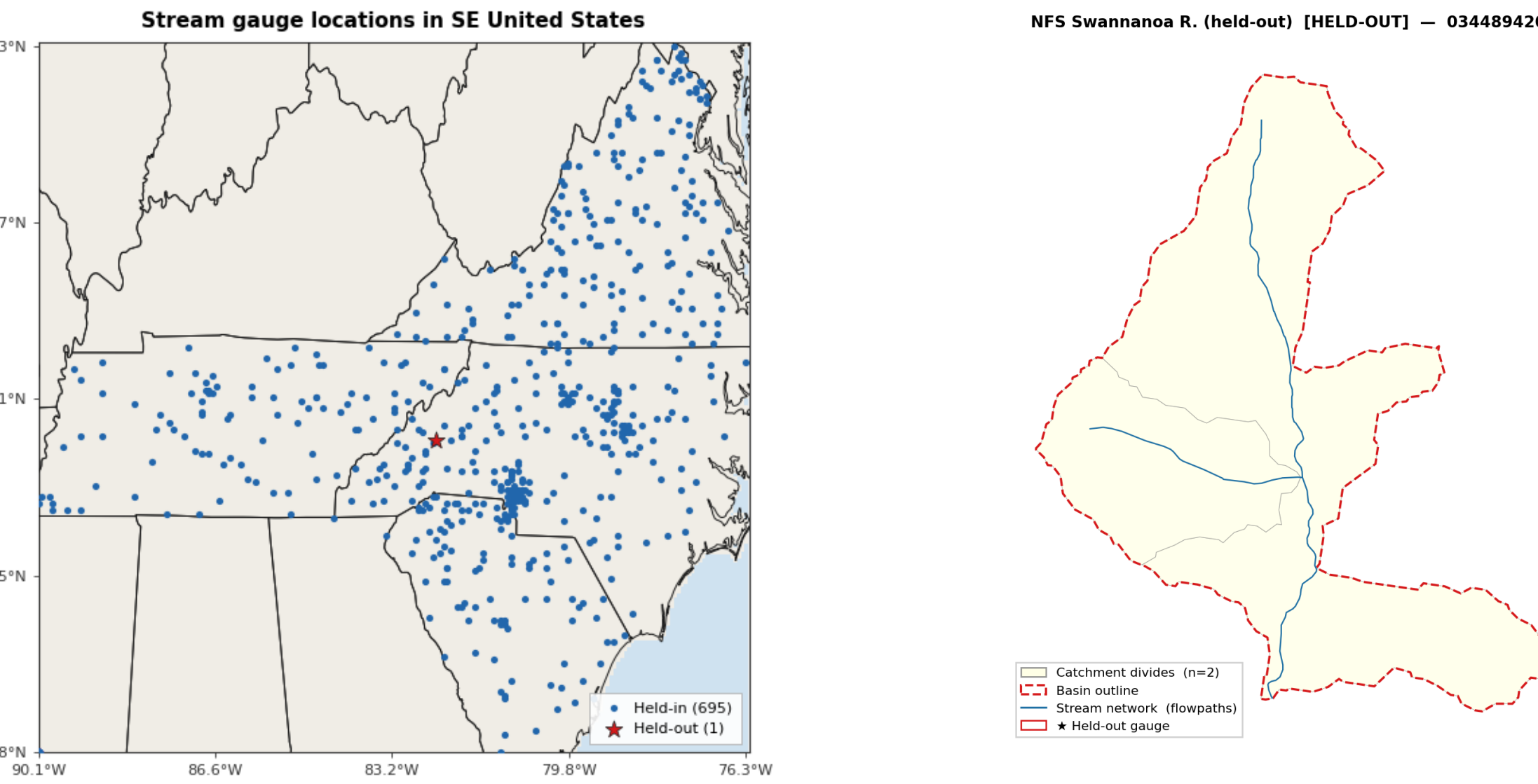


Figure 5 Locations of stream gauges and the Nextgen hydrofabric of North Fork Swannanoa River (held out)

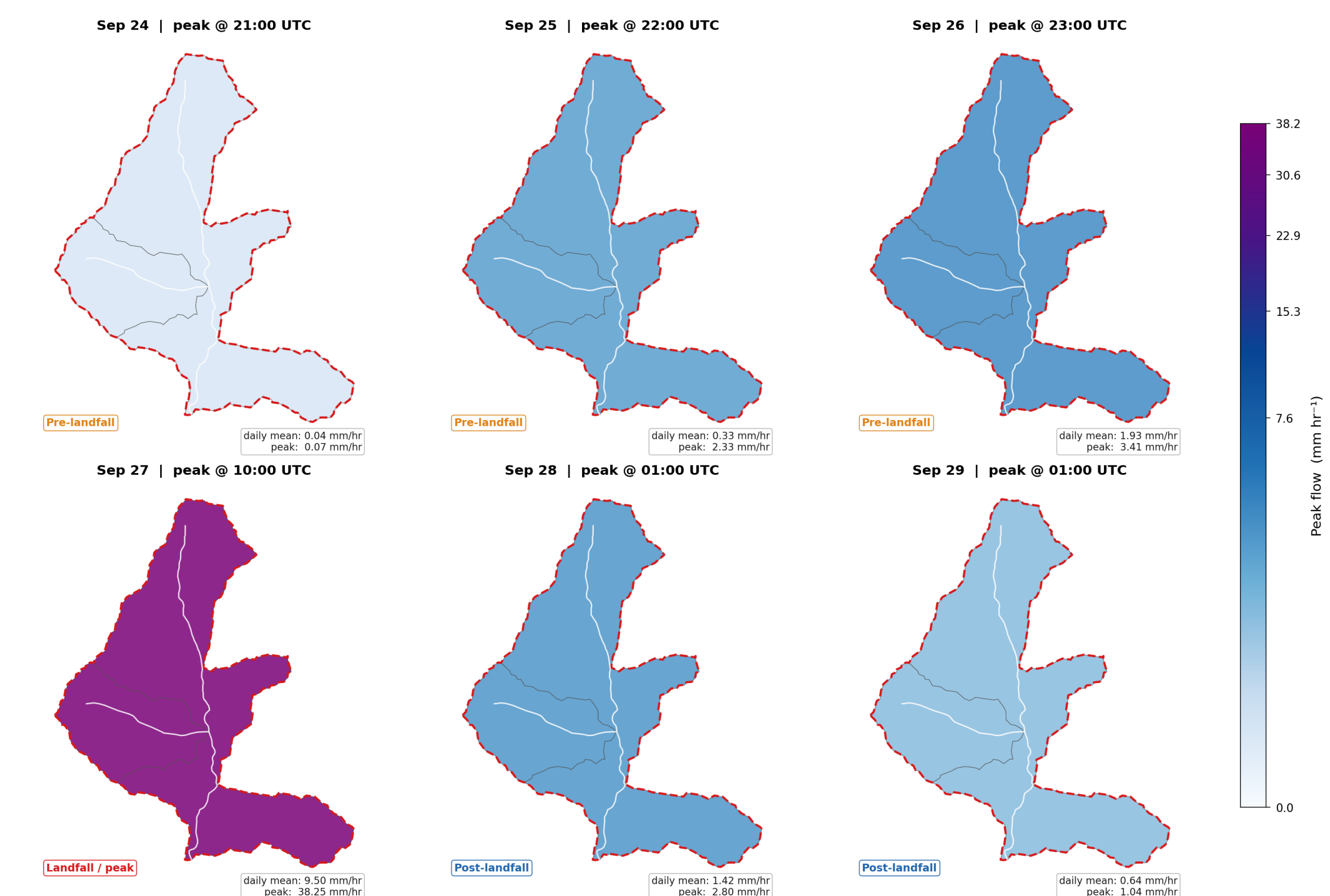


Figure 6 Peak streamflow at the held-out gauge during Hurricane Helene from September 24 to 29, 2024

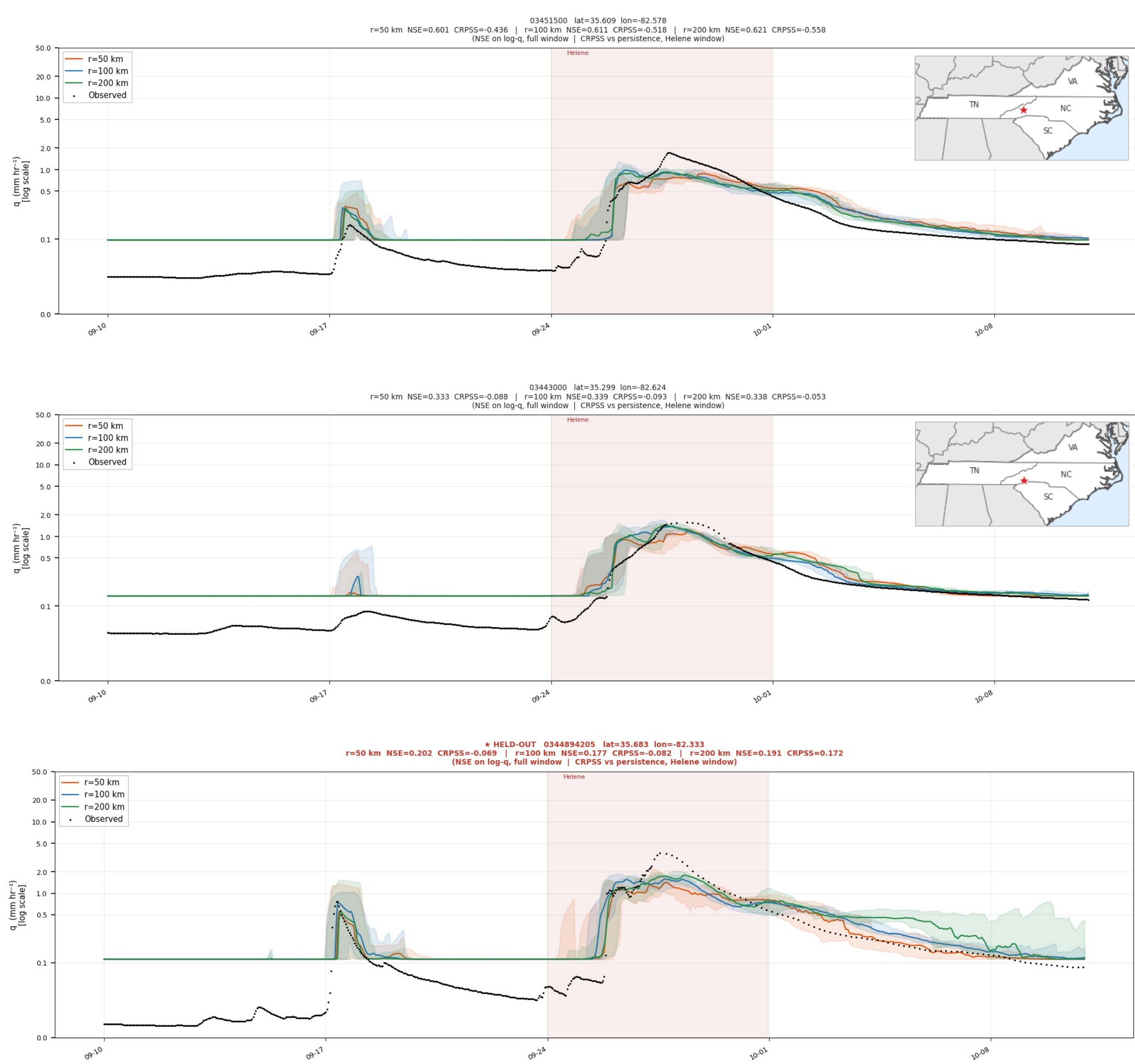


Figure 7 Ensemble hourly streamflow forecasts from a CNN trained on kriged streamflow (ranges: 50 km, 100 km, 200 km) at the French Broad River near Asheville, near Blantyre, and the North Fork Swannanoa River (top to bottom).

Key Findings

- Kriging provides a simple, scalable method for spatially complete reconstruction at daily and hourly timescales
- Neural network forecasts trained on hourly kriged streamflow struggle to capture peak flows at gauged and ungauged locations.

Future work

- Integrate kriged streamflow into Nextgen Water Resources Modeling Framework (Ogden et al., 2026) and evaluate the ensemble forecasts during Helene
- Improve generalization at ungauged locations
- Enhance peak flow prediction for downstream decision support
- Quantify streamflow forecast uncertainty during extreme events



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